

Noncontracting Tails of Unbounded Collatz Orbits: An Exact Reduction in Lean

Collatz proof project

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Abstract

Using the repository's reciprocal-summability theorem, we prove that every unbounded positive natural Collatz orbit has arbitrarily late tails whose multiplicative coefficients never contract. Conversely, every positive seed with no coefficient contraction has an unbounded orbit. Thus nondivergence is equivalent to the assertion that every positive seed has some coefficient contraction. We also prove backward transport and one residue restriction on a least noncontracting seed. Neither side of the global equivalence is established; nontrivial cycles remain a separate problem. No mathematical priority claim is made.

1 Definitions and the tail reduction

Use the shortcut map $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n . Let $x_t = T^t(N)$ and let $j_t(N)$ count odd states before time t . Write

$$a_t(N) = \frac{2^t}{3^{j_t(N)}}.$$

A seed is *noncontracting* when $2^t \leq 3^{j_t(N)}$ for every t , or equivalently $a_t(N) \leq 1$ for every t . This is an infinite condition; a finite prefix check does not establish it.

Theorem 1. *For every $N > 0$,*

$$(\forall B \exists t B < T^t(N)) \iff \exists s \forall t 2^t \leq 3^{j_t(T^s(N))}.$$

On an unbounded orbit, such indices s occur beyond every prescribed time. This does not assert that every sufficiently late index has the property.

From unboundedness to a noncontracting tail

The earlier module `OrbitSummability.lean` proves $a_t(N) \rightarrow 0$ for every unbounded positive orbit. Since $a_0(N) = 1$, all sufficiently late values are less than one. A maximum on the remaining finite initial set is therefore a global maximum. Choose an index s attaining it. The odd-count cocycle gives the exact identity

$$a_{s+t}(N) = a_s(N) a_t(T^s(N)).$$

The left side is at most $a_s(N) > 0$, hence the shifted factor is at most one. This proves the forward implication.

The argument can start from any shifted unbounded orbit. Bounded ideal correction propagates to shifts and forces escape there, so each shift is unbounded. Applying the same maximum argument after any prescribed index gives arbitrarily late noncontracting tails.

From a noncontracting seed to unboundedness

Suppose a positive noncontracting seed had $x_t \leq B$ for every t . The exact correction identity

$$2^t x_t = 3^{j_t} N + d_t$$

would imply $d_t \leq 2^t x_t \leq 3^{j_t} B$. Therefore its ideal corrections $c_t = d_t/3^{j_t}$ would be bounded above by B . The earlier reciprocal theorem proves that bounded ideal correction forces escape to infinity, contradicting the assumed bound on x_t . Thus a positive noncontracting seed is unbounded. If it occurs on another orbit, that orbit is unbounded as well, completing Theorem 1.

2 The precise global obligation

Theorem 2. *The following propositions are equivalent:*

1. *Every positive natural orbit is bounded above.*
2. *For every $N > 0$, there exists t such that $3^{j_t(N)} < 2^t$.*

If (1) holds, the converse in Theorem 1 excludes a noncontracting seed, proving (2). If (2) holds and an unbounded orbit existed, its positive noncontracting tail would contradict (2).

This equivalence concerns nondivergence. It does not rule out nontrivial cycles or prove that bounded orbits reach one. The earlier first-contraction certificate supplies actual descent when a coefficient contraction occurs within 65 steps, but it supplies neither a universal contraction time nor the all-time assertion (2).

3 Backward transport and a residue restriction

A finite prefix of length k with $2^t \leq 3^{j_t(N)}$ for every $t \leq k$, followed by a noncontracting tail at $T^k(N)$, is noncontracting at all times. For $t = k + r$, multiply the inequalities for the prefix and tail, using $j_{k+r}(N) = j_k(N) + j_r(T^k(N))$. This transport rule is proved for arbitrary prefix lengths.

In particular, an odd predecessor of a noncontracting seed is noncontracting: its first coefficient is $3/2$, and the remaining coefficients are that factor times the tail coefficients. A noncontracting seed must itself be odd, since an even first step would have coefficient $1/2$.

Theorem 3. *If N is noncontracting and $N \equiv 2 \pmod{3}$, then*

$$M = \frac{2N - 1}{3}$$

is a smaller positive odd noncontracting seed. Consequently a least positive noncontracting seed cannot be $2 \pmod{3}$.

Indeed N is odd and at least five, so M is an odd positive integer, $M < N$, and $3M + 1 = 2N$. Thus $T(M) = N$ and odd backward transport applies. This excludes the residue only for a least such seed. It does not exclude all noncontracting seeds of that residue, and it leaves residues zero and one modulo three unresolved.

4 Verification and remaining work

Source: `lean/Collatz/NoncontractingTail.lean`. The central declarations are `unbounded_iff_noncontracting_tail`, `unbounded_has_arbitrarily_late_noncontracting_tail`, and `all_orbits_bounded_iff_all_contract`. Backward transport is `neverContracts_of_prefix`; the least-seed restriction is `least_noncontracting_not_two_mod_three`. The module build and selected 139-declaration axiom audit pass using only standard foundations. No new axiom or `sorry` is used. Build with `lake build Collatz.NoncontractingTail` from `lean/`.

The full proof still needs an exclusion of every positive noncontracting seed, and an exclusion of nontrivial cycles. The reduction proves that the noncontracting class covers the escape problem; it does not prove that the class is empty. The Collatz conjecture remains unresolved.